EXPERIMENT 1

AIM: To establish and study the point-to-point optical fiber communication link and calculate propagation loss, attenuation in optical fibers.

SOFTWARE USED: OptiSystem 23

THEORY:

Optical fiber communication is a method of transmitting information from one point to another using light as the carrier signal through an optical fiber. A basic point-to-point optical fiber communication system consists of an optical transmitter, optical fiber, and optical receiver.

The optical transmitter converts the input electrical signal into an optical signal using a light source such as an LED or laser diode. The optical signal is coupled into the optical fiber and propagates through the fiber based on the principle of total internal reflection. At the receiving end, a photodetector converts the received optical signal back into an electrical signal.

During propagation through an optical fiber, the optical power decreases due to absorption, scattering, bending, and other transmission losses. This reduction in optical power is known as propagation loss.

If P_{in} is the optical power at the input of the fiber and P_{out} is the optical power at the output of the fiber, the propagation loss is given by:

$$\text{Propagation Loss} = 10 \log_{10} \left(\frac{P_{in}}{P_{out}} \right) \ dB$$

The attenuation of an optical fiber represents the loss of optical power per unit length of the fiber. If L is the length of the fiber in kilometres, attenuation is given by:

$$\alpha = \frac{10}{L} \log_{10} \left(\frac{P_{in}}{P_{out}} \right) \ dB/km$$

Thus, by measuring the optical power at the input and output of the fiber using optical power meters, the propagation loss and attenuation of the optical fiber can be determined. In this experiment, a point-to-point optical fiber communication link is established and studied using OptiSystem.

PROCEDURE:

1. Open OptiSystem and create a new simulation project.
2. Construct a point-to-point optical fiber communication link consisting of an optical transmitter, optical fiber, and optical receiver as shown in the block diagram
3. Connect an Optical Power Meter at the input of the optical fiber to measure the transmitted optical power P_{in}
4. Connect another Optical Power Meter at the output of the optical fiber to measure the received optical power P_{out}
5. Set the required parameters of the optical transmitter and optical fiber, including the fiber length.
6. Run the simulation.
7. Observe and record the optical power readings from both power meters.
8. Calculate the propagation loss using:

$$\text{Propagation Loss} = 10\log_{10}\left(\frac{P_{in}}{P_{out}}\right)\ dB$$

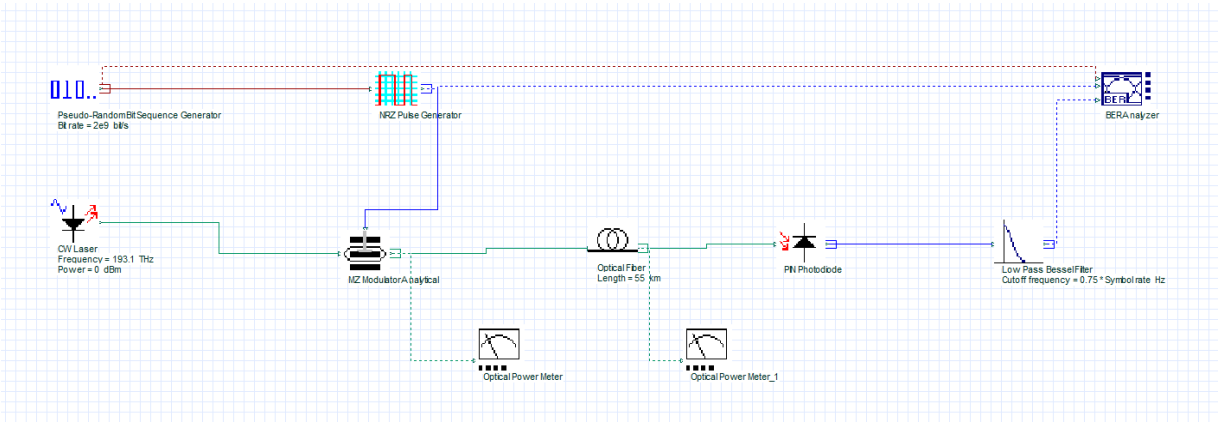
9. Calculate the attenuation of the optical fiber using:

$$\alpha = \frac{10}{L}\log_{10}\left(\frac{P_{in}}{P_{out}}\right)\ dB/km$$

   where L is the length of the optical fiber in kilometres.

10. If required, repeat the simulation for different fiber lengths and record the corresponding power readings, propagation loss, and attenuation.
11. Compare the obtained results and study the effect of fiber length on the propagation loss and attenuation.

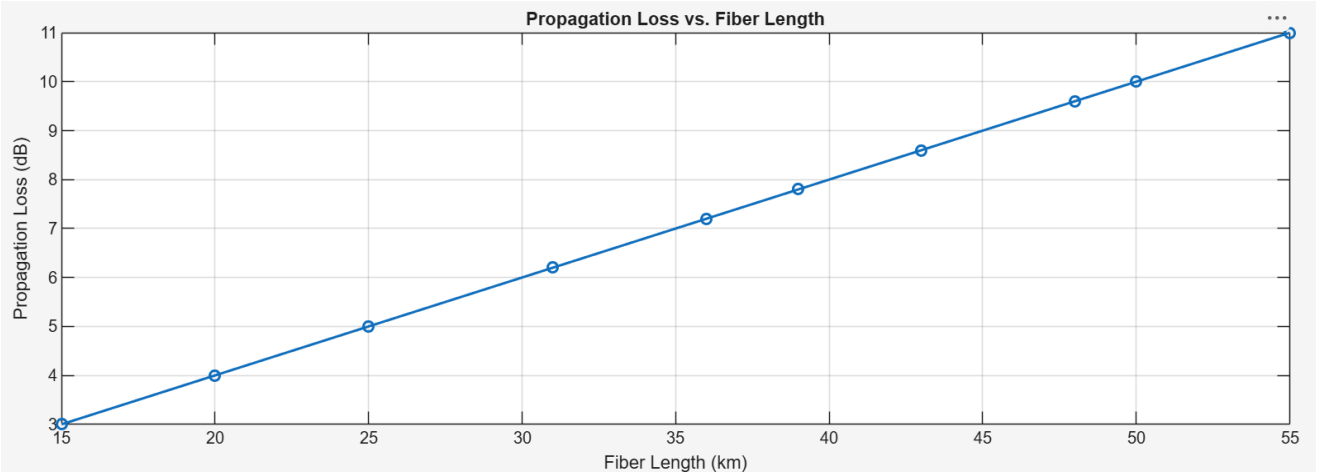
BLOCK DIAGRAM:



OBSERVATIONS:

S. No.	Fiber Length, L (km)	P_in (in µW)	P_out (in µW)	Propagation Loss (dB)	Attenuation (dB/km)
1	15	346.986	173.905	3.00	0.20
2	20	342.113	136.198	4.00	0.20
3	25	351.858	111.268	5.00	0.20
4	31	288.509	69.209	6.20	0.20
5	36	361.605	68.902	7.20	0.20
6	39	288.509	47.880	7.80	0.20
7	43	342.113	47.225	8.60	0.20
8	48	376.223	41.252	9.60	0.20
9	50	361.605	36.159	10.00	0.20
10	55	366.478	29.110	11.00	0.20

GRAPHS:



RESULTS:

The point-to-point optical fiber communication link was successfully established and studied using OptiSystem. The propagation loss of the optical fiber was determined for different fiber lengths and was found to increase with the increase in fiber length. The attenuation of the optical fiber was calculated to be approximately 0.2 dB/km.

EXPERIMENT 2

AIM: To observe working principle and characteristics of LED and plot its P-I characteristics.

SOFTWARE USED: OptiSystem 23

THEORY:

A Light Emitting Diode (LED) is a semiconductor optoelectronic device that converts electrical energy into light. It consists of a p-n junction and emits light through electroluminescence when forward biased. The emitted light's wavelength depends on the semiconductor's band-gap energy.

The V-I characteristic shows the relationship between forward voltage and current. The current increases rapidly after the knee or turn-on voltage. The P-I characteristic represents optical output power versus forward current, where optical power generally increases with current. The P-frequency characteristic shows how optical output power varies with the driving signal frequency, with the response decreasing at high frequencies due to carrier recombination time and limited modulation bandwidth.

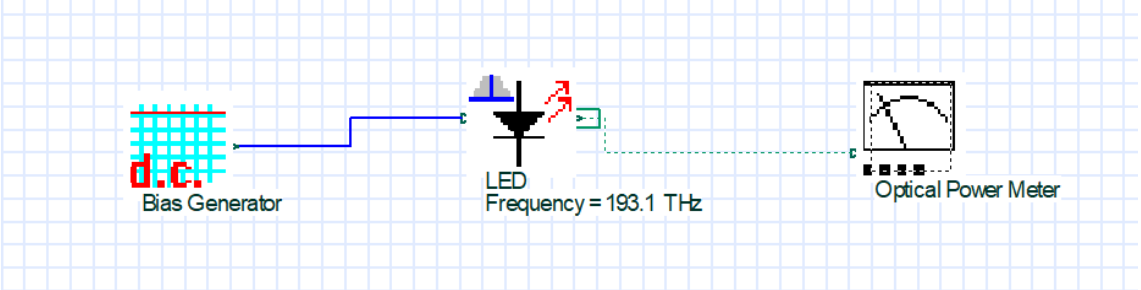
These characteristics can be studied by varying the applied voltage, current, and frequency and measuring the corresponding optical output power using OptiSystem simulation.

PROCEDURE:

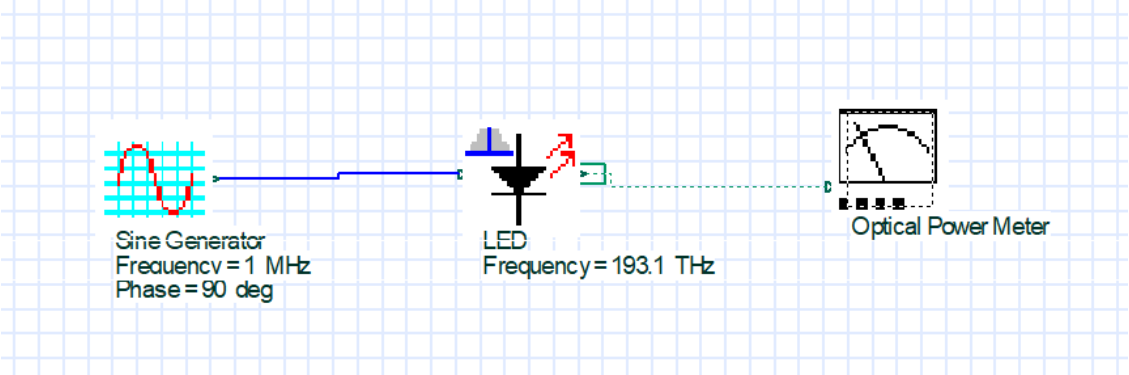
1. Open OptiSystem and create a new simulation project.
2. Connect DC Bias Generator → LED → Optical Power Meter and set the required LED parameters.
3. Vary the DC bias amplitude over the selected range and record the optical power.
4. Calculate the effective current using $I = A^2/2$, where A is the bias amplitude, and plot P versus I to obtain the P–I characteristic.
5. Use the appropriate electrical measurement setup to obtain the LED voltage and current, and plot I versus V to obtain the V–I characteristic.
6. For the P–f characteristic, apply a sinusoidal electrical signal to the LED and sweep its frequency from 1 MHz to 1 GHz, keeping the signal amplitude and bias conditions constant.
7. Record the corresponding optical power at each frequency using the Optical Power Meter and normalize it with respect to the power at 1 MHz, P / P_0.
8. Plot Optical Power (P) or normalized power (P/P_0) versus Frequency (f) and analyze the frequency response of the LED.

BLOCK DIAGRAM:

For P vs I characteristics:



For P vs Frequency characteristics:



OBSERVATIONS:

For P vs I characteristics:

S. No.	Amplitude (a.u.)	Effective Current I = A²/2 (mA)	Optical Power P (μW)
1	0.045	1.013	506.486
2	0.080	3.200	1603.0
3	0.110	6.050	3029.0
4	0.140	9.800	4896.0
5	0.180	16.200	8132.0
6	0.220	24.200	12113.0
7	0.260	33.800	16896.0
8	0.290	42.050	21091.0
9	0.320	51.200	25648.0

For P vs I characteristics:

S. No.	Frequency (in MHz)	Power (µW)	Normalized Power (P/P_0) (P_0 = 4913 µW at 1 MHz)
1	1	4913	1.000
2	10	1956	0.398
3	50	1447	0.295
4	100	792.815	0.161
5	150	454.528	0.0925
6	200	282.847	0.0576
7	300	128.823	0.0262
8	500	49.478	0.0101
9	700	25.182	0.00513
10	1000	12.609	0.00257

CALCULATIONS:

P-I Characteristic — Derivation

The DC Bias Generator's Amplitude parameter (a.u.) does not directly represent the LED drive current. Through calibration trials (Amplitude = 0.01, 0.02, 0.015, 0.05), it was found that the LED's optical output follows:

$P = \zeta \times I$, where $I = (\text{Amplitude})^2 / 2$

with $\zeta = 0.5$ W/A (the LED's set Slope efficiency). This indicates the Amplitude parameter behaves as a normalized signal amplitude, and the effective drive current is obtained by squaring and halving it — consistent with how power/current is derived from a signal amplitude in this simulation convention.

Sample calculation (Amplitude = 0.045):

$$I = (0.045)^2 / 2 = 0.001013\,A = 1.013\,mA$$

This conversion was applied to all 9 data points to obtain the effective current column shown in Observations.

Verification of slope:

$Slope \ = \ \Delta P \ / \ \Delta I \ = (25648 \ - \ 506.486) \ \mu W/(51.2 \ - \ 1.013) mA \approx 500 \ \mu W/mA \ = \ 0.5 \ W/A$

This matches the LED's configured Slope efficiency exactly, confirming the derived current values are physically consistent with the component's own parameters.

P-f Characteristic — Derivation

Reference power at low frequency (1 MHz, unaffected by roll-off): P_0 = 1986 µW
−3 dB power level: $P_0 \ / \ 2 \ = \ 993 \ \mu W$

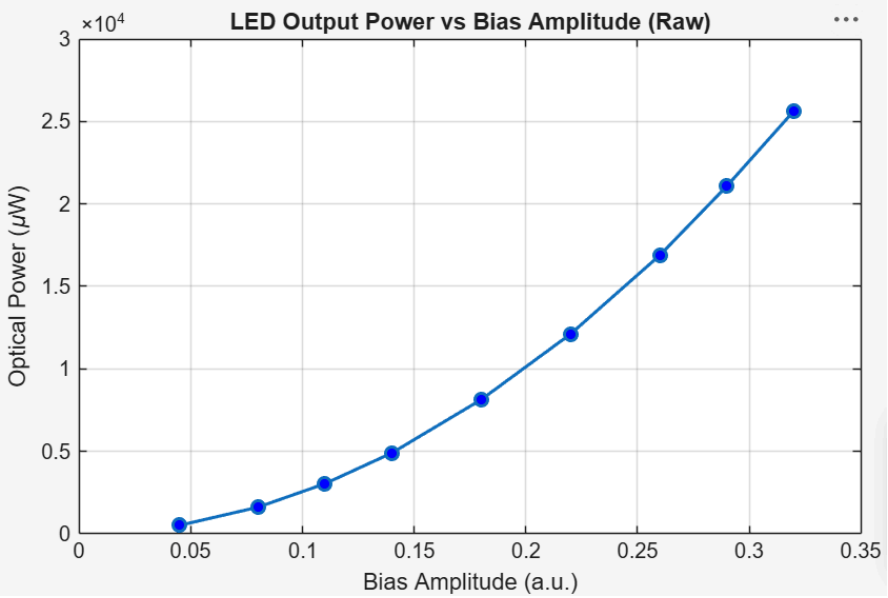
This falls between the 50 MHz (1423 µW) and 100 MHz (792.815 µW) readings. Using log-linear interpolation:

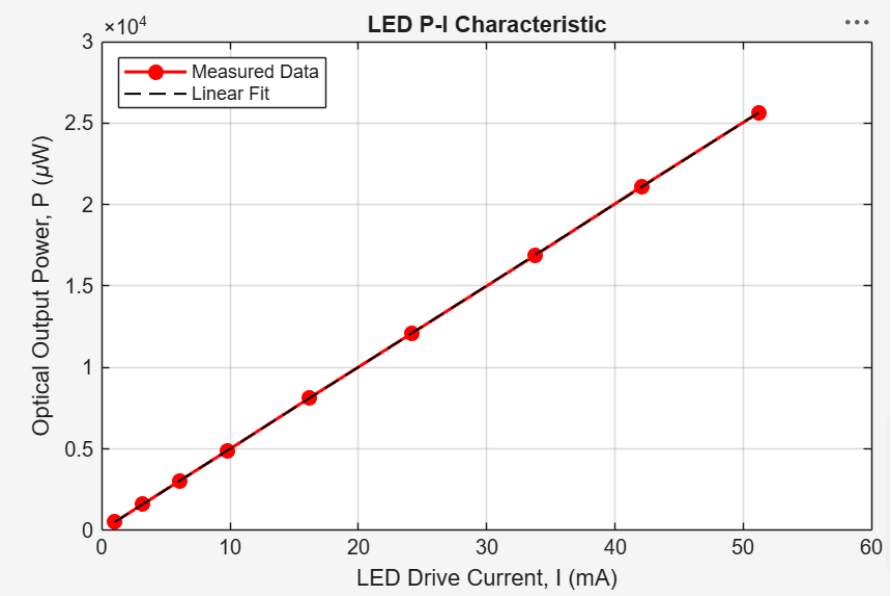
$$f \ = \ 50 \times 10^{[0.301 \times (1423-993)/(1423-792.8)]} \approx 80 \ MHz$$

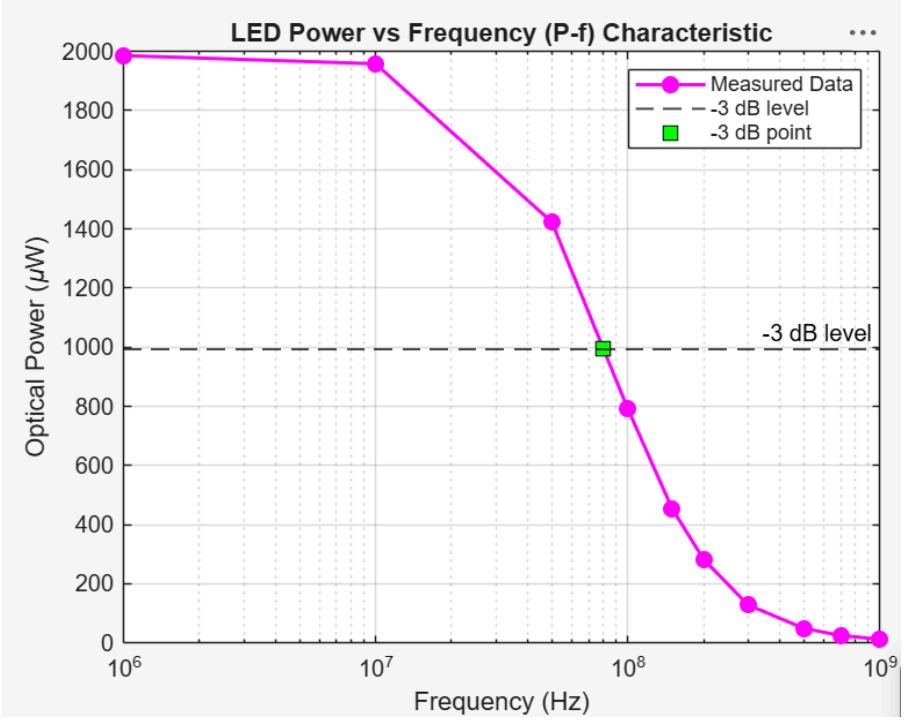
Theoretical single-pole estimate from electron lifetime (τ = 1 ns):
f_c = 1/(2πτ) ≈ 159 MHz

The measured bandwidth is lower than this single-pole estimate because the LED model combines **two** cascaded time constants (electron lifetime and RC constant, both 1 ns) rather than one — two cascaded first-order roll-offs produce a steeper combined response and a lower effective −3dB point than either alone.

GRAPHS:







RESULTS:

1. The P-I characteristic obtained using OptiSystem was linear, with a slope efficiency of approximately 0.5 W/A, matching the configured value.
2. The P-f characteristic remained nearly flat up to 10 MHz, followed by a gradual decrease in optical output power.
3. The measured 3 dB bandwidth of the LED was approximately 80 MHz.
4. The theoretical bandwidth for an electron lifetime of 1 ns is approximately 159 MHz. The lower simulated bandwidth is attributed to the combined effects of the electron lifetime and RC time constant.

EXPERIMENT 3

AIM: To observe working principle and characteristics of LASER and plot its P-I characteristics.

SOFTWARE USED: OptiSystem 23

THEORY:

A LASER (Light Amplification by Stimulated Emission of Radiation) is an optoelectronic device that produces a highly coherent, monochromatic, and directional beam of light. A semiconductor laser consists of a forward-biased semiconductor junction placed between two reflecting facets, forming an optical resonant cavity.

When the injection current is applied, electrons and holes are injected into the active region and recombine to produce photons. At sufficiently high current, stimulated emission becomes dominant and optical amplification occurs. The minimum current required to initiate significant laser action is called the threshold current I_{th}.

The P–I characteristic represents optical output power versus injection current. Below the threshold current, the optical output is very small and mainly due to spontaneous emission. Above threshold, the optical power increases approximately linearly with injection current.

The V–I characteristic represents the relationship between the applied voltage and injection current of the laser diode. As the applied voltage increases beyond the diode's turn-on region, the current increases rapidly.

The P–f characteristic represents the variation of optical output power with modulation frequency. The optical power generally decreases at higher frequencies due to the finite carrier lifetime and parasitic effects of the laser diode, indicating its limited modulation bandwidth.

In the Fabry–Perot laser, the two cleaved semiconductor facets act as partially reflecting mirrors, forming the resonant cavity. When the optical gain becomes sufficient to overcome cavity losses, laser oscillation is established and coherent optical radiation is produced.

PROCEDURE:

A. P-I Characteristic

1. Place a Fabry Perot Laser component. Connect Optical Null to its Input port and Electrical Null to its Modulation Input.
2. Connect the Output to an Optical Power Meter (set to Total Power).

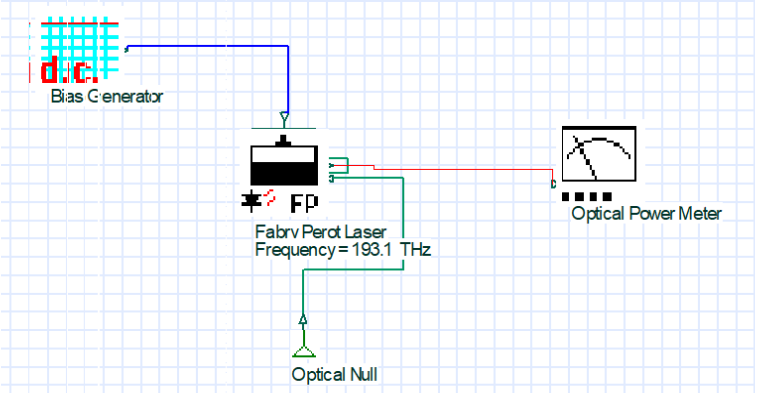
3. In the laser's properties, ensure Enabled = True and Parameterized = Normal. Note the auto-calculated Threshold current (45.58 mA).
4. Keep Modulation peak current = 0. Enable Sweep Mode on Bias current, sweeping values below and above threshold (10–90 mA).
5. Run the simulation and record the Power Meter reading at each bias current.

B. Power-Frequency Characteristic

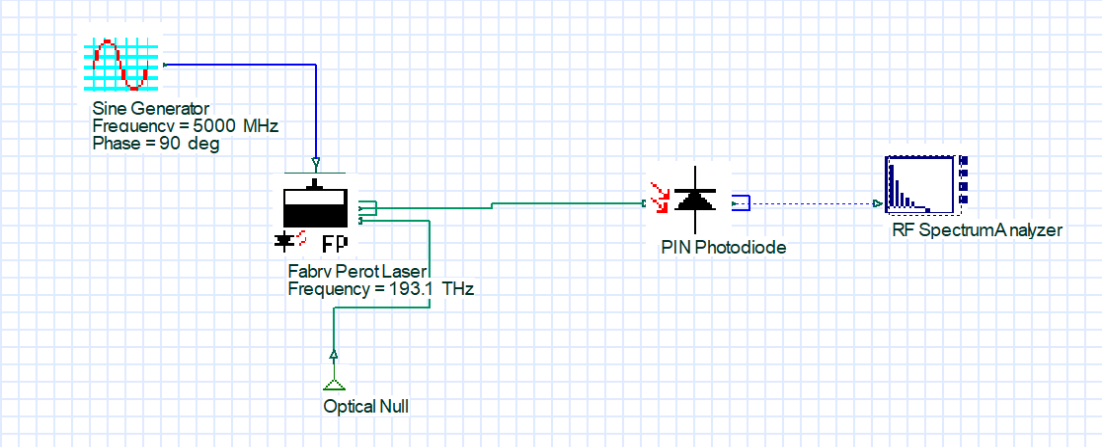
1. Replace the Electrical Null with a Sine Wave Generator. Fix Bias current = 90 mA, Modulation peak current = 10 mA, Sine Amplitude = 1 (a.u.).
2. Update Layout Parameters (Bit rate = 40 Gbps, Sequence length = 16384) to ensure sufficient cycles at low sweep frequencies.
3. Replace the Power Meter with a Photodetector PIN → RF Spectrum Analyzer.
4. Enable Sweep Mode on the Sine Generator's frequency (100 MHz–10 GHz). For each run, center the Spectrum Analyzer on the test frequency with a fixed amplitude scale (0 to −100 dBm).
5. Record the peak power (dBm) at the modulation frequency for each sweep point.

<u>BLOCK DIAGRAM</u>:

A. P-I Characteristic:



B. Power-Frequency Characteristic



OBSERVATION TABLES:

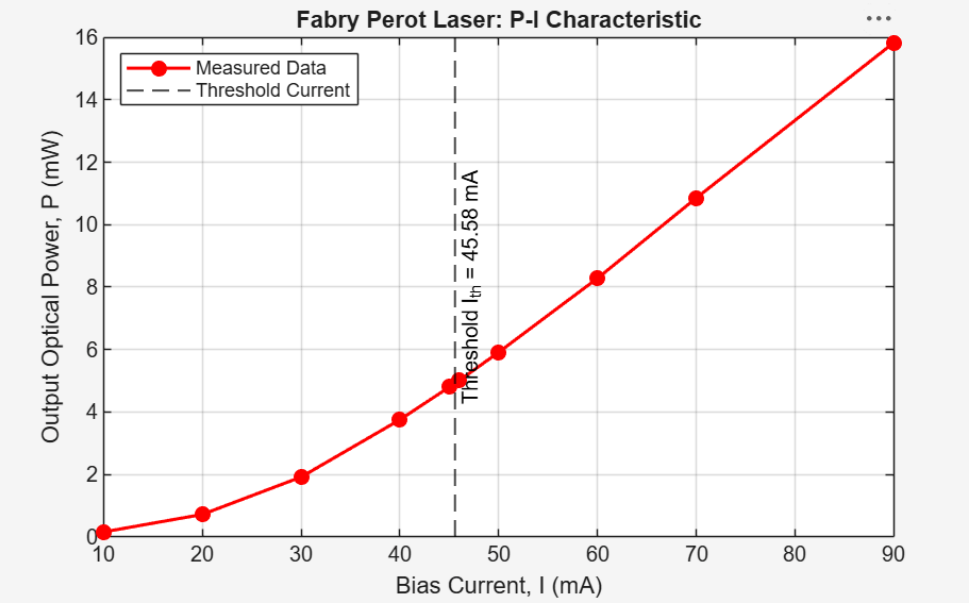
Table A — P-I Characteristic

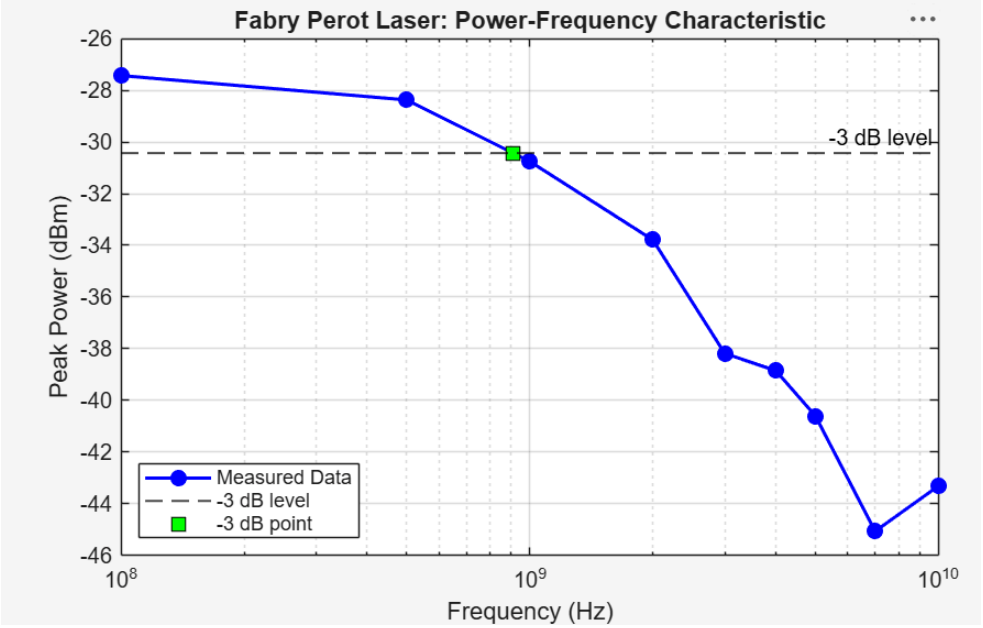
S. No.	Bias Current (mA)	Output Power (mW)
1	10	0.156396
2	20	0.717653
3	30	1.916
4	40	3.751
5	45	4.797
6	46	5.003
7	50	5.913
8	60	8.280
9	70	10.853
10	90	15.828

Table B — Power-Frequency Characteristic

S. No.	Frequency	Peak Power (dBm)
1	100 MHz	-27.430
2	500 MHz	-28.374
3	1 GHz	-30.753
4	2 GHz	-33.788
5	3 GHz	-38.183
6	4 GHz	-38.848
7	5 GHz	-40.601
8	7 GHz	-45.064
9	10 GHz	-43.312

GRAPHS:





CALCULATIONS:

1. Threshold Current

The threshold current for the Fabry Perot Laser was auto-calculated by OptiSystem from the physical/gain parameters of the device:

I_{th} = 45.58 mA

From the P-I data, the slope (ΔP/ΔI) was found to increase gradually through this region, confirming the transition from spontaneous-emission-dominated to stimulated-emission-dominated output near the calculated threshold.

2. Power-Frequency: −3dB Bandwidth (measured)

Reference power (at 100 MHz): $P_0 = -27.430$ dBm
−3dB power level: $P_0 - 3 = -30.430$ dBm

This level falls between the 500 MHz (−28.374 dBm) and 1 GHz (−30.753 dBm) readings. Using log-linear interpolation:

Fraction = $(P_1 - P_{3dB}) / (P_1 - P_2)$ = (−28.374 − (−30.430)) / (−28.374 − (−30.753)) = 2.056 / 2.379 ≈ 0.864

$\log(f) = \log(500\times10^6) + 0.864 \times [\log(1\times10^9) - \log(500\times10^6)]$
$\log(f) = 8.699 + 0.864 \times 0.301 \approx 8.959$

f_{3dB} (measured) ≈ 10^8.959 ≈ 910 MHz

3. Theoretical Relaxation Oscillation Frequency

Using the standard rate-equation expression:

$$f_r = (1/2\pi) \times \sqrt{\frac{v_g \times a \times \eta \times (I - I_{th})}{(q \times V)}}$$

With: $v_g = 8.57\times10^9$ cm/s, $a = 1\times10^{-16}$ cm², $\eta = 0.4$, $I - I_{th} = 44.42$ mA, $V = 1.8\times10^{-10}$ cm³, $q = 1.602\times10^{-19}$ C:

$f_r \approx 3.66$ GHz

Corresponding theoretical 3dB bandwidth (using $f_{3dB} \approx 1.55 \times f_r$ for a lightly damped second-order response):

f_{3dB} (theoretical) ≈ 5.7 GHz

RESULTS:

The P-I characteristic showed output power increasing with bias current, with a clear change in slope near the calculated threshold current of 45.58 mA. The measured 3dB bandwidth from the P-f characteristic was found to be approximately 910 MHz, notably lower than the theoretical estimate of ~5.7 GHz, indicating strong damping effects in the laser's response.

EXPERIMENT 4

AIM: To study and plot the refractive index profile and numerical aperture of optical fiber using optical simulation tools/MATLAB.

SOFTWARE USED: MATLAB

THEORY:

Optical fiber consists of a core surrounded by a cladding, with the core having a higher refractive index than the cladding. The variation of refractive index across the core determines the type of optical fiber and its light-guiding properties.

The two important refractive-index profiles are:

- **Step-index fiber:** The refractive index is constant throughout the core and changes abruptly at the core-cladding boundary.
- **Graded-index fiber:** The refractive index gradually decreases from the center of the core toward the core-cladding boundary.

For a graded-index fiber, the refractive index distribution can be represented by

$$n(r) = n_1 \left[1 - 2\Delta \left(\frac{r}{a}\right)^{\alpha}\right]^{1/2}, \qquad 0 \le r \le a$$

where:

- n_1 = refractive index at the center of the core
- n_2 = refractive index of the cladding
- a = core radius
- r = radial distance from the fiber axis
- Δ = relative refractive-index difference
- α = profile parameter that determines the shape of the refractive-index profile.

The relative refractive-index difference is

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

The value of α controls the transition between different refractive-index profiles. For a parabolic graded-index profile, $\alpha = 2$. As α is increased, the profile becomes increasingly

steep near the core-cladding boundary. In the limiting case of a very large α, the graded-index profile approaches a step-index profile.

Thus, by varying α in MATLAB or an optical simulation tool, the transition from a smooth graded-index profile to an abrupt step-index profile can be observed.

The numerical aperture (NA) represents the light-gathering ability of an optical fiber. For a step-index fiber,

$$NA = \sqrt{n_1^2 - n_2^2}$$

and for a fiber in air, it is related to the maximum acceptance angle θ_a by

$$NA = \sin\theta_a$$

where θ_a is the maximum angle at which light can enter the fiber and still be guided by total internal reflection.

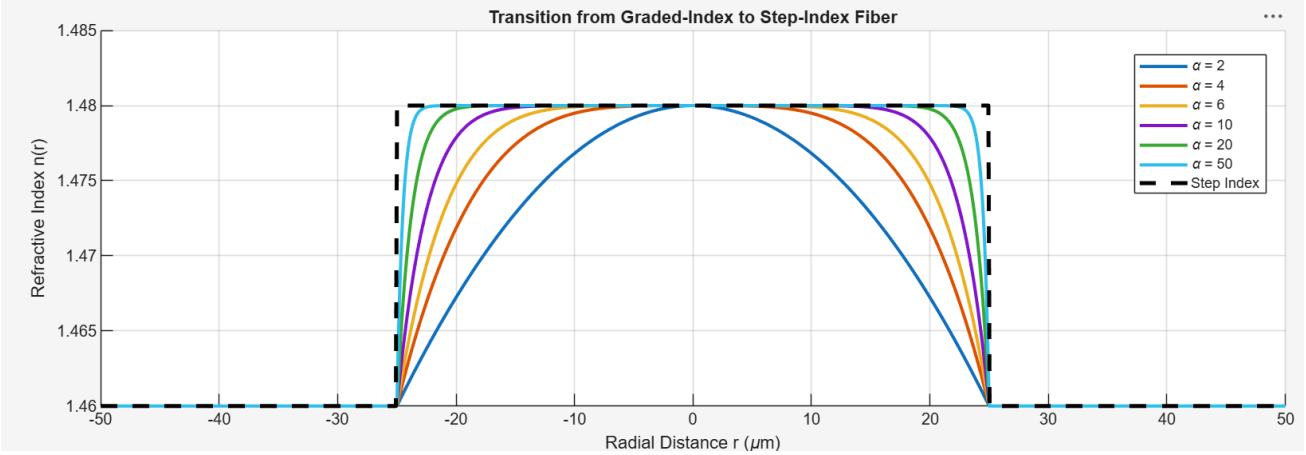
The refractive-index profile and numerical aperture can be studied and plotted using MATLAB or optical simulation software by specifying the core and cladding refractive indices and varying the profile parameter α.

MATLAB INPUT:

```
clc;
clear;
close all;
% Fiber parameters
n1 = 1.48;          % Core refractive index
n2 = 1.46;          % Cladding refractive index
a = 25;             % Core radius (um)
% Relative refractive index difference
Delta = (n1^2 - n2^2)/(2*n1^2);
% Radial distance
r = linspace(-50,50,2000);
% Alpha values
alpha_values = [2 4 6 10 20 50];
figure;
hold on;
for alpha = alpha_values
   n = n2 * ones(size(r));
   core = abs(r) <= a;
   n(core) = n1 * sqrt(1 - 2*Delta*(abs(r(core))/a).^alpha);
   plot(r,n,'LineWidth',2,...
        'DisplayName',['\alpha = ' num2str(alpha)]);
end
```

```
% Step-index profile
n(abs(r) <= a) = n1;
plot(r,n,'k--','LineWidth',2.5,...
    'DisplayName','Step Index');
xlabel('Radial Distance r (\mum)');
ylabel('Refractive Index n(r)');
title('Transition from Graded-Index to Step-Index Fiber');
legend('Location','best');
grid on;
xlim([-50 50]);
hold off;
```

OUTPUT:



RESULTS:

The refractive-index profiles of the optical fiber were successfully plotted for different values of the profile parameter α. It was observed that for lower values of α, the refractive index varies gradually from the center of the core toward the cladding, representing a graded-index profile. As α increases, the refractive-index transition becomes increasingly steep at the core-cladding boundary. For very large α, the profile approaches an abrupt transition, representing a step-index profile.

The numerical aperture of the fiber was also determined from the core and cladding refractive indices. For $n_1 = 1.48$ and $n_2 = 1.46$,

$$NA = \sqrt{n_1^2 - n_2^2} \approx 0.2425$$

Thus, the numerical aperture of the optical fiber is approximately 0.243.